

Disadvantages of digital power supply

Despite their numerous advantages, digital power supplies come with certain limitations. The loop

bandwidth is constrained by the computing power of the microcontroller and the overhead tasks it handles. Additionally, the resolution of the PWM module limits the accuracy of feedback loop regulation.

Constraints in ADC resolution and sampling rates further affect performance, while the overall capability of the microcontroller (CPU) remains a critical factor in determining the system's effectiveness.

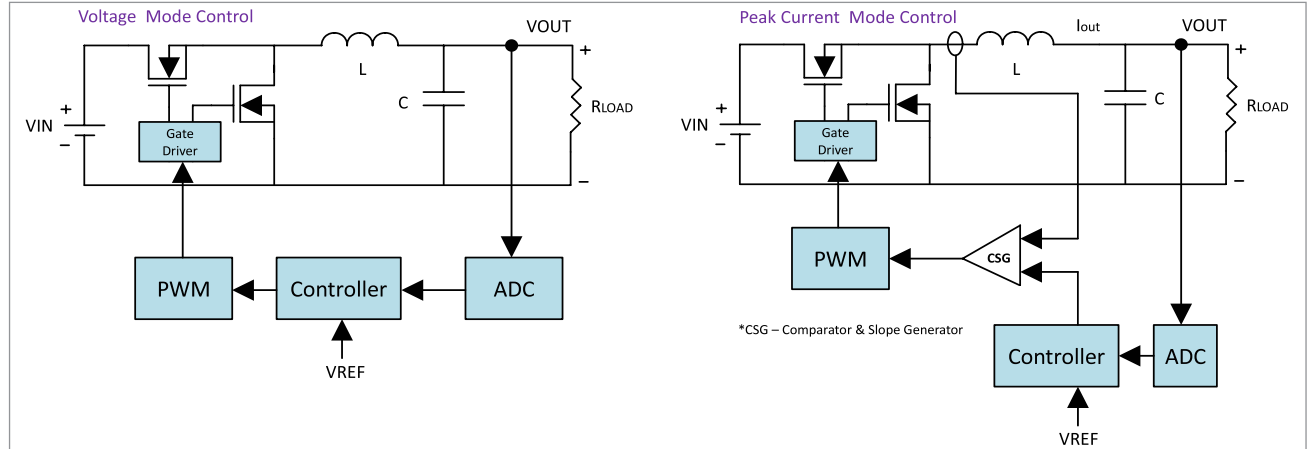


Fig. 3: Two types of switching regulators: (L) voltage mode control and (R) peak current mode control (Credit: www.infineon.com)

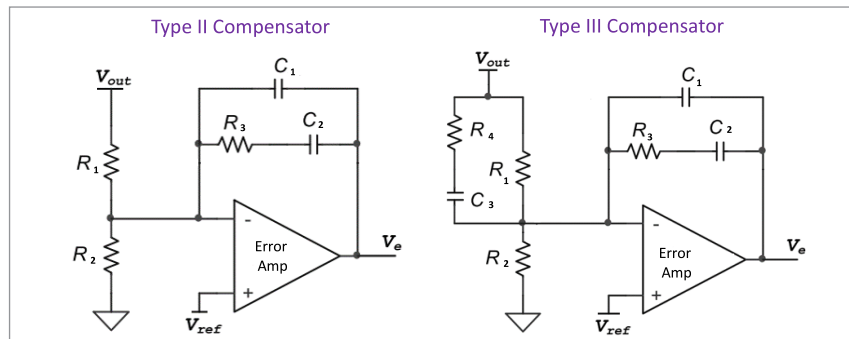


Fig. 4: Types of compensators. (L) Type II compensator (R) Type III compensator (Credit: www.infineon.com)

These challenges underline the need for careful design and component selection in digital power supply systems.

Control algorithms in switching regulators

Typically, switching regulators have two types of control algorithms; each has advantages and disadvantages.

Voltage mode control (VMC).

- Fixed switching frequency

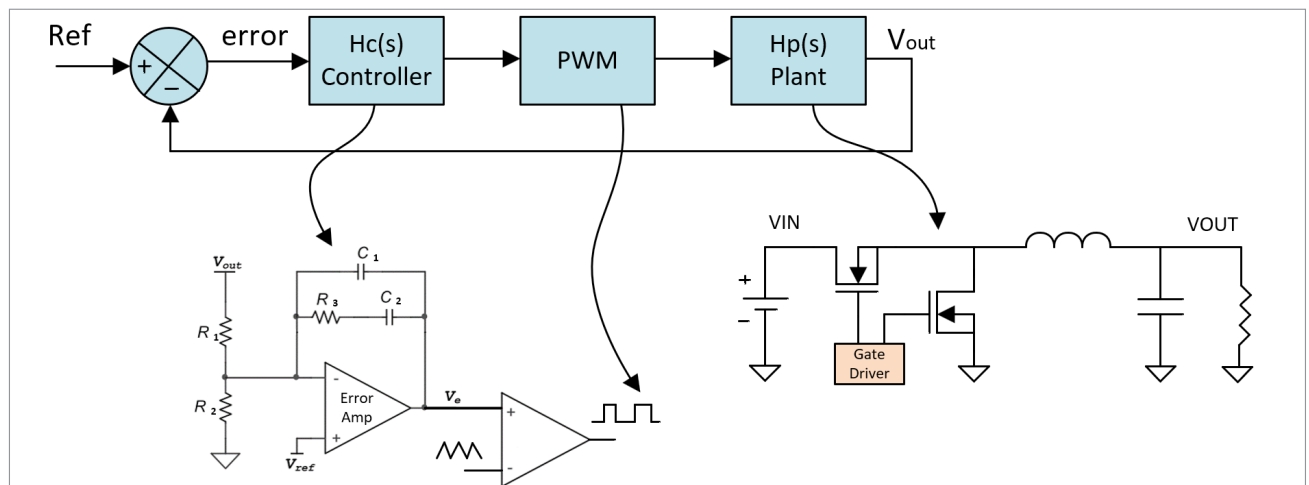


Fig. 5: Voltage mode control in analogue power supply system schematic (Credit: Biricha Digital)